

Description of Invisible Spectral Flicker (ISF) Device

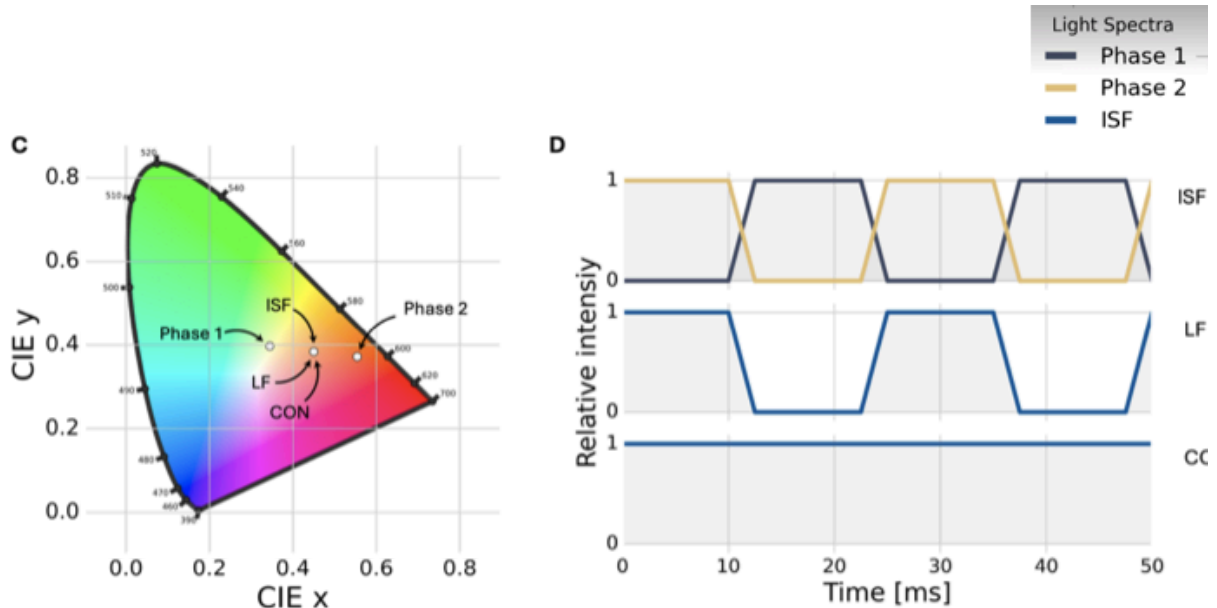


Fig. 1.

Figure 1 depicts a technical description of a non-invasive neurostimulation device (device depicted in figure 2) developed by the Danish company OptoCeutics, and represents an innovative light-based therapy system designed to deliver non-invasive brain entrainment of lower gamma-band activity through precisely modulated luminance flicker. As detailed in the figure 1, the device uses two specific colors within the CIE 1931 color space to generate two distinct phases of light—Phase 1 and Phase 2—that alternate at a 40 Hz frequency. The alteration between these two colors is what creates the core mechanism of invisible spectral flicker (ISF), enabling neural gamma entrainment without inducing perceptible visual discomfort, distinguishing it from conventional luminance-based stroboscopic flicker therapies.

The device employs a dual-modality flicker system, comprising luminance flicker (LF) and invisible spectral flicker (ISF), as illustrated in Panel (D) of Figure 1. The luminance flicker (LF) involves binary “ON-OFF” alterations, which, if unmasked, manifest as a stroboscopic effect, potentially leading to adverse reactions such as dizziness, migraines, nausea, or even triggering

photosensitive epilepsy. However, the device masks this LF through the synchronous alternation between Phase 1 and Phase 2 of the ISF.

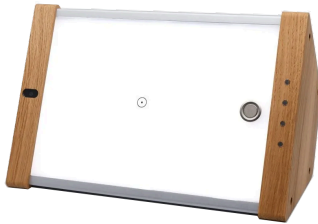


Fig. 2.

Panel (C) of Figure 1 maps these phases as closely positioned chromaticity coordinates within the CIE 1931 color space, representing two subtly differentiated spectral compositions—essentially two highly similar shades of white light. These phases are engineered to be metamERICALLY matched, meaning they appear visually identical to the human eye under typical viewing conditions, yet their spectral profiles differ sufficiently to induce the desired 40 Hz modulation. As the light oscillates between these phases over time (depicted in milliseconds across three distinct operational regimes: constant color [CC], LF, and ISF), the perceived brightness remains stable. This creates an "almost imperceptible flicker effect," where the overt luminance changes are concealed by the spectral shifts, allowing the 40 Hz rhythm to penetrate the visual system and entrain neural oscillations in the brain's gamma band (around 40 Hz).

The technology employs high-precision LED emitters, optimized by nanotechnology specialists to ensure the transitions occur seamlessly at 40 cycles per second—equivalent to 40 flashes per second. The relative intensity waveform in Panel (D) visualizes this: the ISF traces (in blue and orange) overlay the LF (in darker blue), demonstrating how the spectral alternation neutralizes visible pulsing while preserving the therapeutic frequency.